

# Raghav Senthil Kumar

Phoenix, AZ | 623-332-2168 | [raghavs9@illinois.edu](mailto:raghavs9@illinois.edu) | [linkedin.com/in/raghav-senthil-kumar](https://linkedin.com/in/raghav-senthil-kumar) | [github.com/raghav](https://github.com/raghav)

## EDUCATION

### University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, Agentic AI, BuildIllinois, Innovation LLC, Agentic AI

## EXPERIENCE

### Founding Software Engineer | Clusion

September 2025 – January 2026

- Built the document search engine (*Python FastAPI*) for a competitive high school debate research SaaS platform hosted on *AWS*, developed by an *Agile* team of **10 engineers** and drawing **10K+ site visitors**
- Indexed **600K+ debate-evidence documents** on *Elasticsearch* via *FastAPI*, returning results in **under 1 second**
- Designed and built a text-retrieval algorithm that extracts evidence segments from debate cases by validating source URLs and filtering statistical outliers with **IQR** on the word-count distribution of tag, citation and evidence text
- Developed a de-duplication algorithm that flagged **30%** of evidence documents as identical, reducing index bloat
- Implemented recency-based relevance so evidence from the current month's debate topic ranks first in results

### Product Development Intern | Jetson

January 2025 – February 2025

- Conducted **7 iterations of UI/UX testing** across 3 experimental features on the Jetson App (100K+ users)
- Delivered a **5-page report** recommending strategies to optimize user engagement directly to Jetson's CEO
- Redesigned user onboarding on *Figma*, cutting it from **12 to 5 pages**, validated via **A/B testing** on 30 test users

### Data Analytics Intern | STEM E

June 2024 – July 2024

- Analyzed 3 years of TikTok data (**200+ posts**) with *Python (pandas)*, *Hootsuite*, *Power BI* and *Microsoft Excel*
- Identified that "science fun facts" posts comprised 69.3% of views, despite accounting for 7% of all content
- Pitched a "fun facts" strategy to the Executive Director, who adopted it; engagement rose **24% (8K+ views)**

## PROJECTS

### Truck Pricing VLM | [Live Website](#) | [Source Code](#)

September 2026

- Led team of **4** to build a pricing algorithm for **Kamion (YC S22)**. **2nd of 7 at UIUC Founder's Hackathon**
- Scraped **15,000** truck-listing images for *data mining*, auditing quality via de-duplication and non-USD filtering
- Designed a *JSON schema* for a **Claude Sonnet 4.6 agent** to extract condition, brand, model and year from photos
- Revised the schema after each wrong prediction until **10 consecutive images** were correct on each of **3 features**
- Built a *quantile regression* pricing model on the extracted features, reaching **87% accuracy** on **30 samples**

### SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Engineered a **suite of 3 edge AI tools** that collects data regarding a school's energy, water and food consumption
- Built a *Python FastAPI* backend that ingests live water, waste & energy data from all 3 tools, runs **4 Random Forest models** to detect anomalies & deliver actionable steps via **Gemma Voice LLM** on a *Next.js* dashboard
- Led a team of 5 across project ideation, development and presentation. **Finalist at USAII Global Hackathon**

### Citation Chain | [Live Website](#) | [Source Code](#)

June 2026

- Built an interactive citation graph (*React Flow*) that maps a paper's outbound citations across **26 research fields**
- Built a dual-API ingestion pipeline resolving papers via *OpenAlex* & backfilling metadata from *Semantic Scholar*
- Deployed a **RAG agent (Claude Sonnet 4.5)** that retrieves session-scoped vector documents (*Supernormality*)
- Stored paper data on *Supabase (PostgreSQL)* and wrote *SQL* queries to filter papers by field, year and citation count

### EcoBin | Research Paper | [Source Code](#)

March 2026 – May 2026

- Designed a two-stage *EfficientNetV2-S* neural network (**4.39M** parameters) in *PyTorch*, trained on **15K+ images** to classify waste into 4 disposal pathways at **96.13% accuracy**. Published a paper on *arXiv* and 2 datasets on *Kaggle*
- Built a synthetic data generation pipeline using *U2-Net* segmentation and alpha-compositing to generate 9K images to train the model to detect contamination in recyclable waste. **Placed 2nd at the Arizona State Science Fair**
- Engineered a smart bin prototype using *Arduino Uno* that sorts the waste item with **less than 2 second latency**

### Chronos | [Live Website](#) | [Source Code](#)

December 2025

- Built a **scheduling agent (React/Firebase)** that auto-books meetings for large groups (30+ ppl) on Google Calendar

- Designed a **5-step *Python* scheduling algorithm** that scans calendar availability, returns a ranked list of time slots and sends email confirmation for each booking. Validated algorithm across **14 unit and integration test cases**
- Created a lobby system that enables the host to authenticate invited participants and queue follow-up meetings
- Licensed Chronos to **8 startups** & signed **2 LOIs worth \$3.5K**. **Won 2025 Congressional App Challenge**

## TECHNICAL SKILLS

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**Programming Languages:** Java, Python, C++, JavaScript, SQL, HTML/CSS

**Frameworks/Platforms:** FastAPI, Flask, React, React Flow, Next.js, Node.js, PostgreSQL, MySQL, Elasticsearch, Supabase, Firebase, AWS, Google Cloud, Vercel, Google Calendar API, OpenAlex API, Semantic Scholar API

**DevTools:** Git, GitHub, Docker, CI/CD, VS Code, Claude Code, Codex, Figma, Bun, Arduino IDE, Kaggle

**AI/ML:** PyTorch, TensorFlow, Keras, scikit-learn, OpenCV, Grad-CAM, Hugging Face, Claude API, Gemma 4, Vercel AI SDK, Supermemory

**Data & Analytics:** pandas, NumPy, Power BI, Microsoft Excel, Hootsuite, A/B Testing

**Concepts/Practices:** Object-Oriented Programming, REST APIs, API Development, Full-Stack Development, Unit Testing, Authentication, Agile, Data Pipelines, Data Ingestion, ETL, Data Quality, Data Validation, Data Cleaning, Data Mining, Data Visualization, Statistical Analysis, LLMs, RAG, Generative AI, Prompt Engineering, AI Agents

## CERTIFICATIONS

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**Harvard CS50x:** Introduction to Computer Science  
**HackerRank Problem Solving (Basic)** Certificate

November 2024  
August 2026